I. TABLICA Laplasove transformacije

Redni broj	$F(s)$	$f(t), \quad t>0$
1.	1	$\delta(t)$ jedinična impulsna funkcija u $t=0$
2.	$\frac{1}{s}$	1 ili $h(t)$ jedinična odskočna funkcija u $t=0$
3.	$\frac{1}{s^2}$	$th(t)$ nagibna funkcija
4.	$\frac{1}{s^n}$	$\frac{1}{(n-1)!}t^{n-1}$ n je prirodan broj
5.	$\frac{1}{s}e^{-as}$	$h(t-a)$ jedinična odskočna funkcija u $t=a$
6.	$\frac{1}{s}(1-e^{-as})$	$h(t)-h(t-a)$ pravougaoni signal
7.	$\frac{1}{s+a}$	e^{-at} eksponencijalna prigušna funkcija
8.	$\frac{1}{(s+a)^n}$	$\frac{1}{(n-1)!}t^{n-1}e^{-at}$ n je prirodan broj
9.	$\frac{1}{s(s+a)}$	$\frac{1}{a}(1-e^{-at})$
10.	$\frac{1}{s(s+a)(s+b)}$	$\frac{1}{ab}\left(1-\frac{b}{b-a}e^{-at}+\frac{a}{b-a}e^{-bt}\right)$
11.	$\frac{s+\alpha}{s(s+a)(s+b)}$	$\frac{1}{ab}\left[\alpha-\frac{b(\alpha-a)}{b-a}e^{-at}+\frac{a(\alpha-b)}{b-a}e^{-bt}\right]$
12.	$\frac{1}{(s+a)(s+b)}$	$\frac{1}{b-a}(e^{-at}-e^{-bt})$
13.	$\frac{s}{(s+a)(s+b)}$	$\frac{1}{a-b}(ae^{-at}-be^{-bt})$

I. TABLICA **Laplasove transformacije (nastavak)**

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14.	$\frac{s+\alpha}{(s+a)(s+b)}$	$\frac{1}{b-a}[(\alpha-a)e^{-at}-(\alpha-b)e^{-bt}]$
15.	$\frac{1}{(s+a)(s+b)(s+c)}$	$\frac{e^{-at}}{(b-a)(c-a)}+\frac{e^{-bt}}{(c-b)(a-b)}+\frac{e^{-ct}}{(a-c)(b-c)}$
16.	$\frac{s+\alpha}{(s+a)(s+b)(s+c)}$	$\frac{(\alpha-a)e^{-at}}{(b-a)(c-a)}+\frac{(\alpha-b)e^{-bt}}{(c-b)(a-b)}+\frac{(\alpha-c)e^{-ct}}{(a-c)(b-c)}$
17.	$\frac{\omega}{s^2+\omega^2}$	$\sin \omega t$
18.	$\frac{s}{s^2+\omega^2}$	$\cos \omega t$
19.	$\frac{s+\alpha}{s^2+\omega^2}$	$\frac{\sqrt{\alpha^2+\omega^2}}{\omega}\sin(\omega t+\phi) \qquad \phi=\operatorname{arctg}\frac{\omega}{\alpha}$
20.	$\frac{s\sin\theta+\omega\cos\theta}{s^2+\omega^2}$	$\sin(\omega t+\theta)$
21.	$\frac{1}{s(s^2+\omega^2)}$	$\frac{1}{\omega^2}(1-\cos\omega t)$
22.	$\frac{s+\alpha}{s(s^2+\omega^2)}$	$\frac{\alpha}{\omega^2}-\frac{\sqrt{\alpha^2+\omega^2}}{\omega^2}\cos(\omega t+\phi) \qquad \phi=\operatorname{arctg}\frac{\omega}{\alpha}$
23.	$\frac{1}{(s+a)(s^2+\omega^2)}$	$\frac{e^{-at}}{a^2+\omega^2}+\frac{1}{\omega\sqrt{a^2+\omega^2}}\sin(\omega t-\phi) \qquad \phi=\operatorname{arctg}\frac{\omega}{a}$
24.	$\frac{1}{(s+a)^2+b^2}$	$\frac{1}{b}e^{-at}\sin bt$
25.	$\frac{1}{s^2+2\zeta\omega_n s+\omega_n^2}$	$\frac{1}{\omega_n\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin\omega_n\sqrt{1-\zeta^2}\,t$
26.	$\frac{s+a}{(s+a)^2+b^2}$	$e^{-at}\cos bt$
27.	$\frac{s+\alpha}{(s+a)^2+b^2}$	$\frac{\sqrt{(\alpha-a)^2+b^2}}{b}e^{-at}\sin(bt+\phi) \qquad \phi=\operatorname{arctg}\frac{b}{\alpha-a}$
28.	$\frac{1}{s[(s+a)^2+b^2]}$	$\frac{1}{a^2+b^2}+\frac{1}{b\sqrt{a^2+b^2}}e^{-at}\sin(bt-\phi) \qquad \phi=\operatorname{arctg}\frac{b}{-a}$

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29.	$\dfrac{1}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$	$\dfrac{1}{\omega^2}-\dfrac{1}{\omega_n\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_n\sqrt{1-\zeta^2}\,t+\phi)$ $\phi=\arccos\zeta$
30.	$\dfrac{s+\alpha}{s[(s+a)^2+b^2]}$	$\dfrac{\alpha}{a^2+b^2}+\dfrac{1}{b}\sqrt{\dfrac{(\alpha-a)^2+b^2}{a^2+b^2}}\,e^{-at}\sin(bt+\phi)$ $\phi=\operatorname{arctg}\dfrac{b}{\alpha-a}-\operatorname{arctg}\dfrac{b}{-a}$
31.	$\dfrac{1}{(s+c)[(s+a)^2+b^2]}$	$\dfrac{e^{-ct}}{(c-a)^2+b^2}+\dfrac{e^{-at}\sin(bt-\phi)}{b\sqrt{(c-a)^2+b^2}}$ $\quad\phi=\operatorname{arctg}\dfrac{b}{c-a}$
32.	$\dfrac{1}{s(s+c)[(s+a)^2+b^2]}$	$\dfrac{1}{c(a^2+b^2)}-\dfrac{e^{-ct}}{c[(c-a)^2+b^2]}$ $+\dfrac{e^{-at}\sin(bt-\phi)}{b\sqrt{a^2+b^2}\sqrt{(c-a)^2+b^2}}$ $\phi=\operatorname{arctg}\dfrac{b}{-a}+\operatorname{arctg}\dfrac{b}{c-a}$
33.	$\dfrac{s+\alpha}{s(s+c)[(s+a)^2+b^2]}$	$\dfrac{\alpha}{c(a^2+b^2)}-\dfrac{(c-\alpha)e^{-ct}}{c[(c-a)^2+b^2]}$ $+\dfrac{\sqrt{(\alpha-a)^2+b^2}}{b\sqrt{a^2+b^2}\sqrt{(c-a)^2+b^2}}\,e^{-at}\sin(bt+\phi)$ $\phi=\operatorname{arctg}\dfrac{b}{\alpha-a}-\operatorname{arctg}\dfrac{b}{-a}-\operatorname{arctg}\dfrac{b}{c-a}$
34.	$\dfrac{1}{s^2(s+a)}$	$\dfrac{1}{a^2}(at-1+e^{-at})$
35.	$\dfrac{1}{s(s+a)^2}$	$\dfrac{1}{a^2}(1-e^{-at}-ate^{-at})$
36.	$\dfrac{s+\alpha}{s(s+a)^2}$	$\dfrac{1}{a^2}[\alpha-\alpha e^{-at}+a(a-\alpha)te^{-at}]$
37.	$\dfrac{s^2+\alpha_1 s+\alpha_0}{s(s+a)(s+b)}$	$\dfrac{\alpha_0}{ab}+\dfrac{a^2-\alpha_1 a+\alpha_0}{a(a-b)}e^{-at}-\dfrac{b^2-\alpha_1 b+\alpha_0}{b(a-b)}e^{-bt}$
38.	$\dfrac{s^2+\alpha_1 s+\alpha_0}{s[(s+a)^2+b^2]}$	$\dfrac{\alpha_0}{c^2}+\dfrac{1}{bc}[(a^2-b^2-\alpha_1 a+\alpha_0)^2$ $+b^2(\alpha_1-2a)^2]^{1/2}e^{-at}\sin(bt+\phi)$ $\phi=\operatorname{arctg}\dfrac{b(\alpha_1-2a)}{a^2-b^2-\alpha_1 a+\alpha_0}-\operatorname{arctg}\dfrac{b}{-a}$ $c^2=a^2+b^2$

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39.	$\dfrac{1}{(s^2+\omega^2)\,[(s+a)^2+b^2]}$	$\dfrac{(1/\omega)\sin(\omega t+\phi_1)+(1/b)\,e^{-at}\sin(bt+\phi_2)}{[4a^2\omega^2+(a^2+b^2-\omega^2)^2]^{1/2}}$ $\phi_1=\operatorname{arctg}\dfrac{-2a\omega}{a^2+b^2-\omega^2} \qquad \phi_2=\operatorname{arctg}\dfrac{2ab}{a^2-b^2+\omega^2}$
40.	$\dfrac{s+\alpha}{(s^2+\omega^2)\,[(s+a)^2+b^2]}$	$\dfrac{1}{\omega}\left[\dfrac{\alpha^2+\omega^2}{c}\right]^{1/2}\sin(\omega t+\phi_1)$ $+\dfrac{1}{b}\left[\dfrac{(\alpha-a)^2+b^2}{c}\right]^{1/2}e^{-at}\sin(bt+\phi_2)$ $c=(2a\omega)^2+(a^2+b^2-\omega^2)^2$ $\phi_1=\operatorname{arctg}\dfrac{\omega}{\alpha}-\operatorname{arctg}\dfrac{2a\omega}{a^2+b^2+\omega^2}$ $\phi_2=\operatorname{arctg}\dfrac{b}{\alpha-a}+\operatorname{arctg}\dfrac{2ab}{a^2-b^2+\omega^2}$
41.	$\dfrac{s+\alpha}{s^2\,[(s+a)^2+b^2]}$	$\dfrac{1}{c}\left(\alpha t+1-\dfrac{2\alpha a}{c}\right)+\dfrac{[b^2+(\alpha-a)^2]^{1/2}}{bc}\,e^{-at}\sin(bt+\phi)$ $c=a^2+b^2$ $\phi=2\operatorname{arctg}\left(\dfrac{b}{a}\right)+\operatorname{arctg}\dfrac{b}{\alpha-a}$
42.	$\dfrac{s^2+\alpha_1 s+\alpha_0}{s^2(s+a)(s-b)}$	$\dfrac{\alpha_1+\alpha_0 t}{ab}-\dfrac{\alpha_0(a+b)}{(ab)^2}-\dfrac{1}{a-b}\left(1-\dfrac{\alpha_1}{a}+\dfrac{\alpha_0}{a^2}\right)e^{-at}$ $-\dfrac{1}{a-b}\left(1-\dfrac{\alpha_1}{b}+\dfrac{\alpha_0}{b^2}\right)e^{-bt}$